Project Summary

Urban Bike Rental Demand and Operations Analysis

1. Project Title

Urban Bike Rental Demand and Operations Analysis

2. Business Problem

The bike-sharing company needs to manage bike availability efficiently across different times and operating conditions. Low-demand periods can leave too many bikes unused, while high-demand periods can cause shortages and poor customer experience. This project analyzes hourly Seoul bike-rental data to identify demand patterns and the effects of season, holidays, temperature, and rainfall. The goal is to support data-driven bike allocation and identify when the maximum number of bikes should be made available.

3. Dataset

The Seoul Bike Sharing Demand dataset was analyzed. The cleaned dataset contains **8,760 hourly records** and 14 original variables, including rental count, hour, temperature, humidity, wind speed, visibility, dew point temperature, solar radiation, rainfall, snowfall, season, holiday, and functioning-day status. The project also created Time_Category and Weather_Category fields for analysis.

4. Data Cleaning

The dataset was inspected for structure, data types, missing values, duplicates, and invalid records. The analysis found **0 missing values, 0 duplicate records, and no invalid records requiring removal**. The Date field was converted to a proper datetime type. Appropriate categorical and numerical data types were retained, and time/rainfall categories were created for downstream SQL, Excel, and Power BI analysis.

5. Key Findings

- **Total bike rentals:** 6,172,314; the overall average hourly rental count is **704.60** bikes.
- **Peak rental hour:** 18:00, with an average of **1502.93** rentals per hour.
- **Highest-demand season:** Summer, with 2,283,234 total rentals and an average of 1034.07 rentals per hour.
- **Time of day:** Evening is the strongest time category, averaging **1187.40** rentals per hour.
- **Holiday effect:** average demand is 715.23 on No Holiday days versus 499.76 on Holidays, showing substantially higher usage on non-holidays.
- **Rainfall effect:** No Rain averages **739.31** rentals, while Low Rain averages 176.66 and Heavy Rain 71.06. Demand therefore falls sharply as rainfall becomes heavier.
- **Best season/time combination:** Summer + Evening, averaging **1821.33** rentals per hour.

Top 5 Hours by Average Bike Demand

Rank	Hour	Average Rentals
1	18:00	1502.93
2	19:00	1195.15
3	17:00	1138.51
4	20:00	1068.96
5	21:00	1031.45

6. Business Recommendations

- **Increase bike availability around the evening peak:** prepare additional bikes and redistribute inventory before the **18:00** peak, with particular attention to the Summer season.

- **Use weather-aware allocation:** maintain higher bike availability during dry conditions and avoid excessive redistribution during heavy rainfall, when observed demand is much lower.
- **Adopt demand-based seasonal planning:** use season, time category, holiday status, and weather together when forecasting demand. This can reduce idle inventory in low-demand periods while reducing shortages during high-demand periods.

Conclusion

The analysis shows that bike demand is strongly time-dependent and is also associated with season and weather conditions. The strongest observed pattern is concentrated around the evening period, with the overall peak at **18:00**. Summer is the highest-demand season, while rainfall is associated with a substantial reduction in rentals. These findings can be used in the Power BI dashboard to support operational decisions, improve bike availability, reduce unused inventory, and improve customer service.

Project tools: Python/Pandas • BeautifulSoup • PostgreSQL • Excel • Power BI